8(a)(i)	downwards	**B1**
8(a)(ii)	PQRS and JKLM	**B1**
8(b)	(as charge separates) an electric field is created (between opposite faces)	**B1**
	(maximum value is reached when) electric force (on electron) is equal and opposite to magnetic force (on electron)	**B1**
8(c)	$V_H = BI / ntq$ $= (4.6 \times 10^{-3} \times 6.3 \times 10^{-4}) / (1.3 \times 10^{29} \times 0.10 \times 10^{-3} \times 1.60 \times 10^{-19})$	**C1**
	$= 1.4 \times 10^{-12}$ V	**A1**
8(d)	semiconductors have a (much) smaller value for *n*	**B1**
	V_H for semiconductors is (much) larger so more easily measured	**B1**